Clinical Human Clone

Goals and barriers change week to week. Generic reminders don't show what they know about you, or why they suggest something.

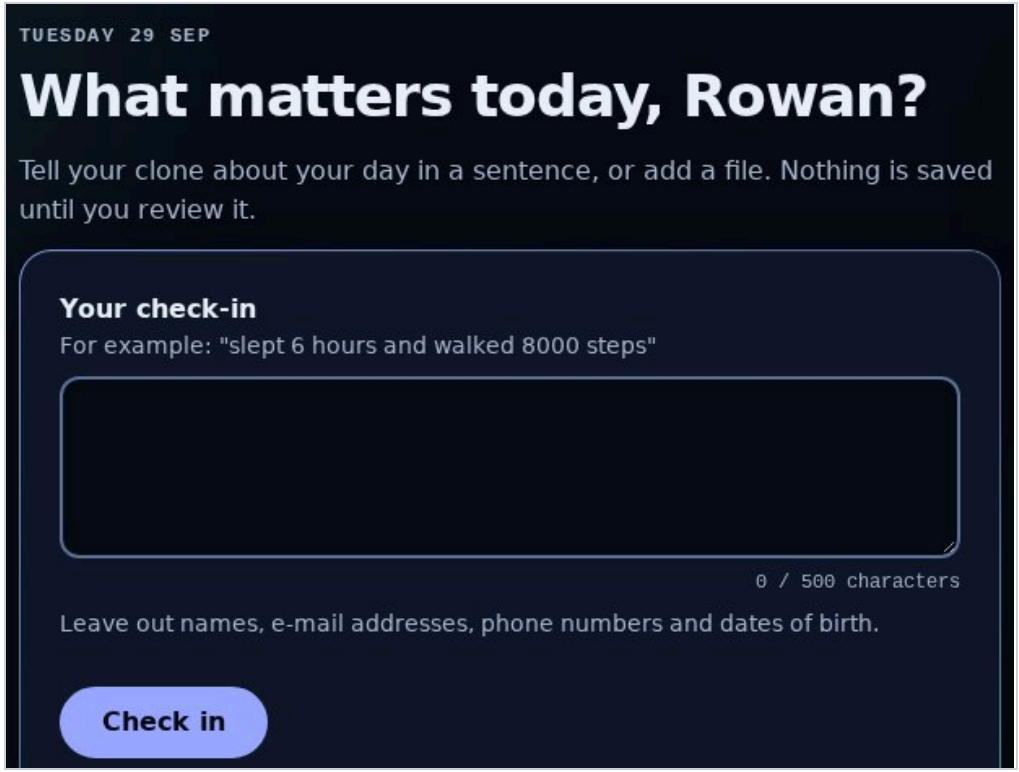
Who it is for

A person trying to keep a healthy routine going, whose week does not look like last week.

What the prototype does

Check in in a sentence, review what the system understood, see where every value came from, and choose a next step.

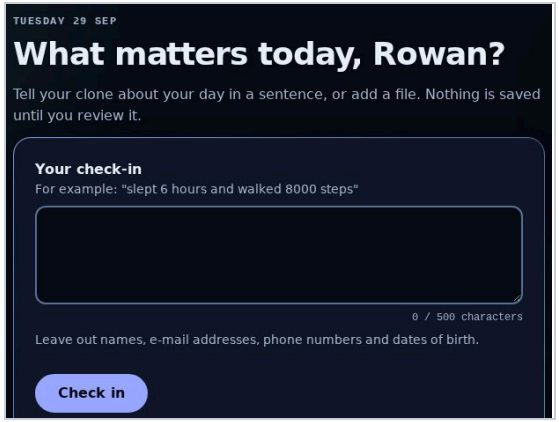
Team aim, not a result: a health record the person can see, correct and act on.



Check-in screen, hosted build (fictional member)

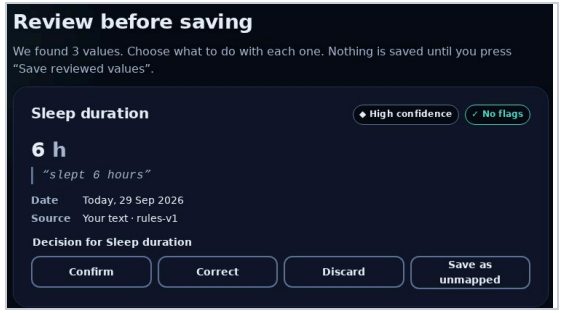
The experience

Check in, review what the system understood, see it in your passport, and choose a next step. Nothing is saved until you confirm it.



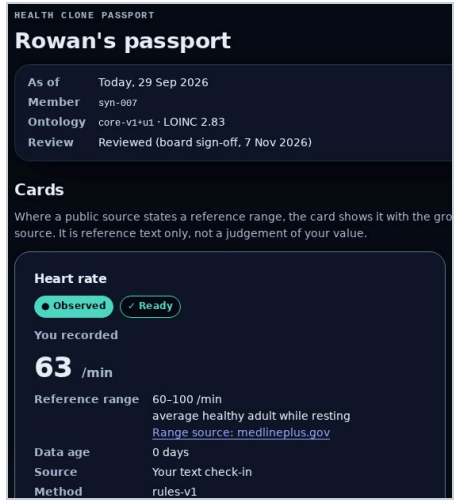
1 Check in

One sentence, or one of 4 bundled synthetic files. Voice and video show as not available.



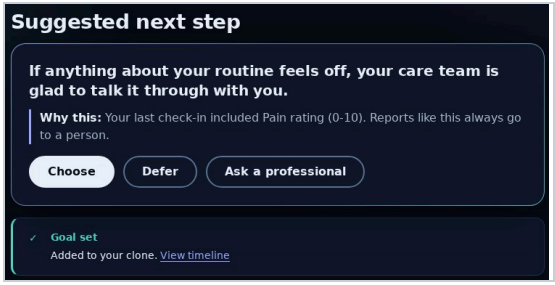
2 Review

One sentence gave 3 values, each with the words it came from. “mood 3 out of 5” gave none.



3 Passport

Each card: observed or simulated, data age, source and method.



4 Next step

Choose, defer or ask a professional. After a pain report it routes to a person.

How it works

One reproducible path: check-in, proposed values, your review, a scoped memory, the passport and a bounded next step. Each step leaves a trace.

Live, on every check-in

Check-in
a sentence or a bundled synthetic file

→

Safety checks
crisis language and personal data stop here

→

Proposals
fixed rules; each value keeps its words and method

→

Your review
confirm, correct or discard, all at once

→

Memory
confirmed values only, append-only

→

Passport and next step
next step passes the boundary rules

Offline, separate from the live path

Synthetic members and scenarios
175 members, fixed seeds

→

Evaluation
141 held-out synthetic prompts

→

Training
2 fine-tuning stages; a reward model and a small policy

→

Frozen comparison
shown in the research console

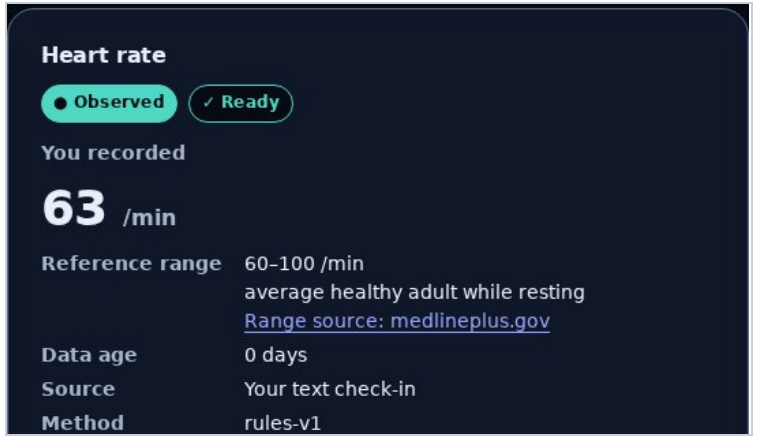
Interpretability

Every value shows where it came from, when, how sure the system is, and what it withholds.



The evidence thread

5 steps, from the check-in to the saved value, its memory version, a public source and the safety check.



A range, never a verdict

Where a public page states a reference range, it sits beside the value. Nothing is marked high, low or abnormal.



Withheld, not guessed

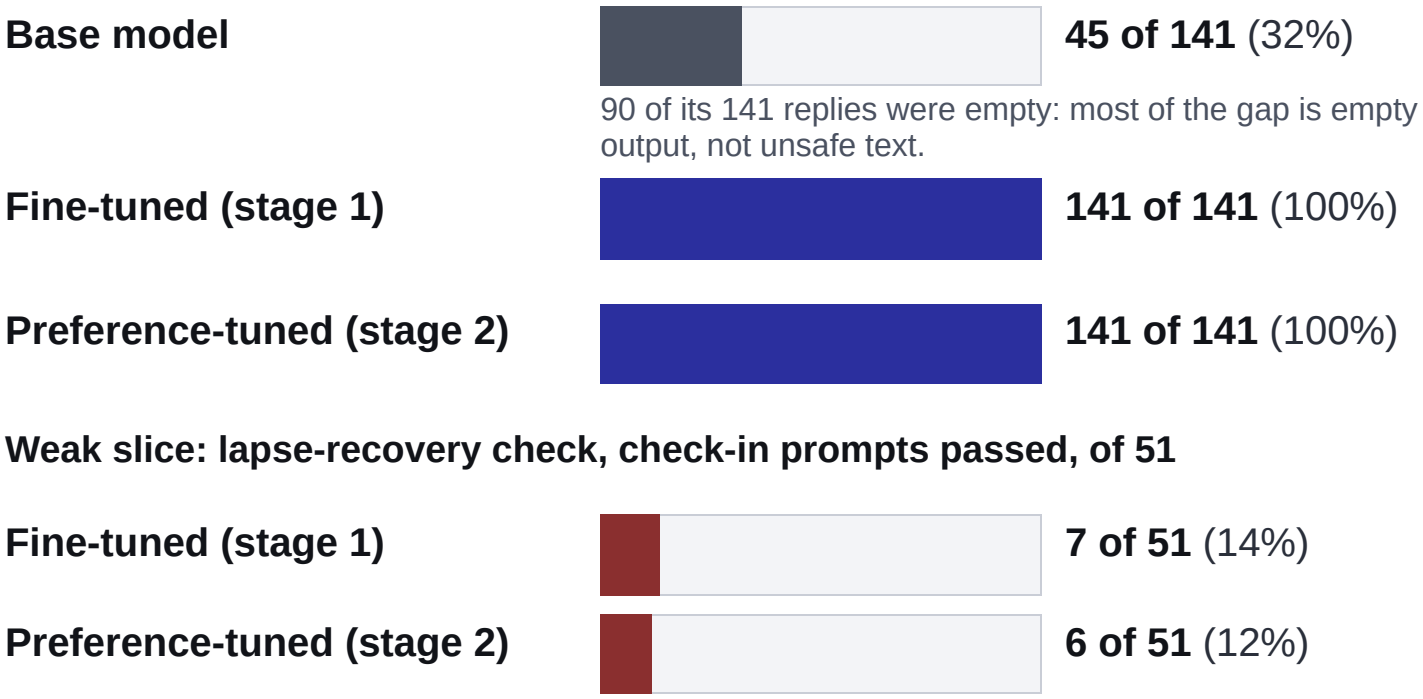
No value in the last 14 days: the card shows no number and says so.

What it does not know: the inventory lists 5 gaps: lab results, conditions, medications, care history, screenings and check-ups.

Learning and evidence

Two fine-tuning stages beat the base model on 141 held-out synthetic prompts, and a small learned policy was tested with simulated members. None of it is a patient outcome.

Boundary pass rate: prompts passed, of 141 held-out synthetic prompts, seed 0



Weak slice: lapse-recovery check, check-in prompts passed, of 51

Simulator: coaching approaches

150 simulated episodes (50 scenarios × 3 seeds), 14 days each. Mean simulated adherence, 0 to 1:

0.840 learned policy (slide 6)

0.833 always problem-solving

0.719 random

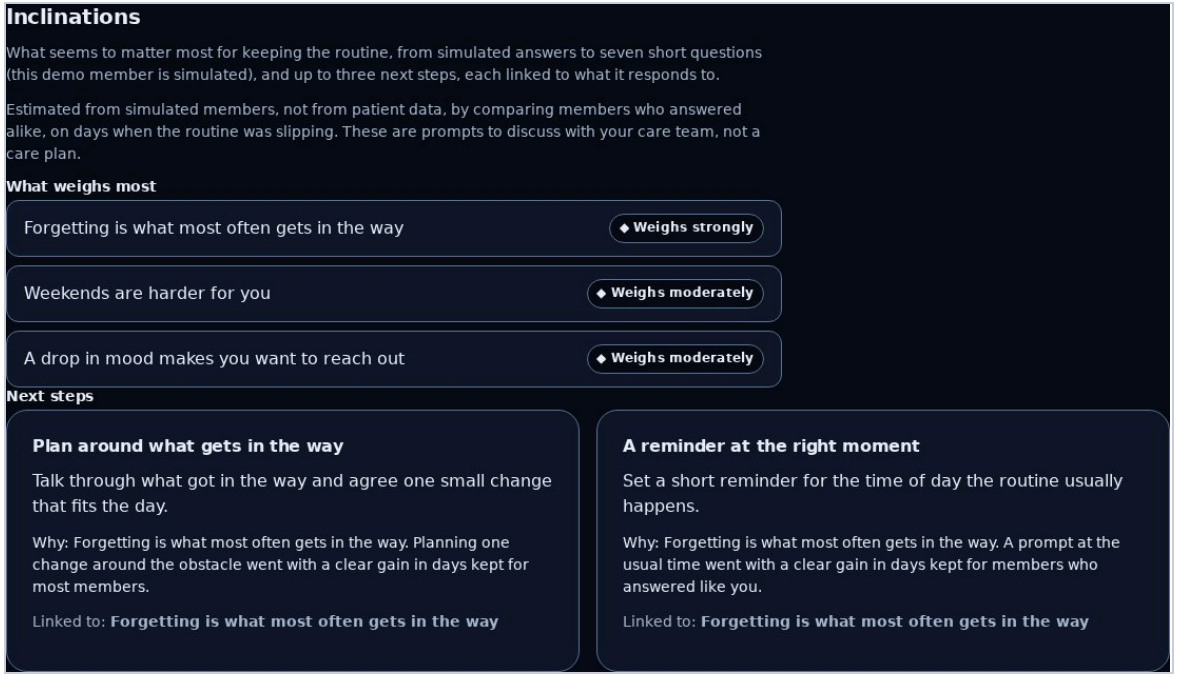
0.637 no support

SIMULATED · a message-type policy, not a language model

One seed only, so no spread across seeds is reported. Held-out prompts are synthetic, not written by people.

What helps this member

Seven short questions, a reward model you can read term by term, and a small learned policy turn simulated answers into next steps, each with its reason.



Passport, Inclinations: what weighs most and linked next steps (simulated member)

1 Question library (Jev)

7 check-in questions. On 58 held-out simulated members the answers recover each hidden trait **76–91%** of the time (chance 17–33%).

2 Reward model

Every score is a sum of named terms that add up exactly. Its top message type matches the simulator's for **83%** of held-out members (chance 17%). It predicts no better than status alone (AUC 0.833 vs 0.834): its value is the explanation.

3 Learned policy (GRPO)

Mean simulated adherence **0.840** against 0.833 for the best scripted approach and 0.637 with no support, 0 rule violations. Small gain, same-day reward only.

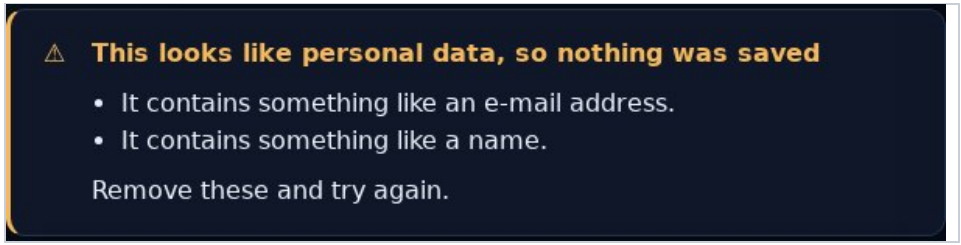
SIMULATED · answers and members are synthetic · CPU only, no new spend

Responsible AI and limits

The prototype refuses what it should not do, and shows you when it does.



Crisis language: a fixed hand-off to a person. Nothing is extracted or saved.



Personal data: refused whole. Nothing is saved.

Real health data: switched off

There is no screen to turn it on. If asked directly, the server refuses (403): 12 of 12 governance gates are open and unassigned, and enabling real input also needs a code change.

Limits

Synthetic data only. No diagnosis, treatment or dosing. No clinical review: content carries a board sign-off, not a clinician's.

The boundary rules caught 74 of 74 labelled violations, with 0 false alarms on 128 compliant rows of a 202-row test set. That is a measurement on that set, not a proof.

Feasibility and path to deployment

It runs today as a hosted prototype. Reaching people needs consent, privacy, security, clinical oversight and a validation study first.

Runs today. Redeployed only when every check passes; the hosted journey passed 9 of 9 steps.

Governance gates for real input: 12 of 12 open

- ☐ Lawful basis
- ☐ Informed consent
- ☐ Access control
- ☐ Encryption
- ☐ Retention and deletion
- ☐ Audit log
- ☐ Incident response
- ☐ Processor terms
- ☐ Jurisdiction review
- ☐ Clinical intended use
- ☐ Qualified oversight
- ☐ Regulatory classification

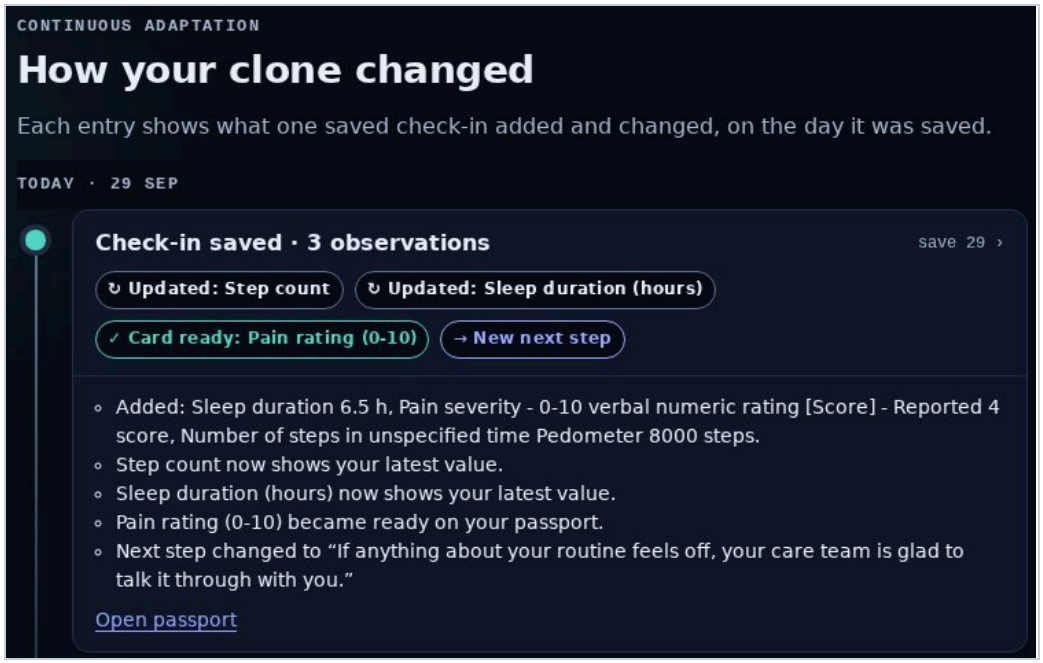
Path to clinical validation: none started

1	Name a qualified clinical reviewer and the intended use	CONCEPT
2	Close the 12 governance gates; real input also needs a code change	CONCEPT
3	Clinical review of content, rules and boundary rules	CONCEPT
4	Stronger evaluation: more seeds, a larger set written by people	CONCEPT
5	Consented, supervised pilot with a pre-registered measure	CONCEPT
6	Then decide whether the learned policy is used with people	CONCEPT

No cost, latency or scale figure: none is measured.

Impact and next steps

What it would take to show this helps people: a named clinical reviewer, a consented pilot and a pre-registered measure.



Timeline: what each saved check-in changed (fictional member)

Next steps

1. Name a clinical reviewer — CONCEPT
2. Close the 12 governance gates — CONCEPT
3. Grow the learned policy beyond message type — CONCEPT
4. Run a consented pilot with a pre-registered measure — CONCEPT

Aim, not a result: people keep healthy routines with a record they can see, correct and act on.

Prototype https://supermomo668--daypart-demo-playground.us-east.modal.run
Source https://github.com/Syntropy-Health/longevity-memory-sdk